

Credit Card Fraud Detection Using Machine Learning

Student Information

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Project Title

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Introduction

This project focuses on identifying fraudulent credit card transactions using machine learning techniques. By analyzing transaction data, the model helps distinguish legitimate activities from suspicious ones, supporting safer digital payment systems.

Objectives

- Identify fraudulent transactions accurately.
- Address class imbalance with the SMOTE technique.
- Build and compare Logistic Regression and Random Forest models.
- Measure performance using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC Score.

Dataset Information

Dataset: Credit Card Fraud Detection Dataset

Total Records: 284,807

Features: 30

Target Variable: Class

0 = Legitimate Transaction

1 = Fraudulent Transaction

Technologies Used

Python

Pandas

NumPy

Matplotlib

Scikit-learn

Imbalanced-learn (SMOTE)